

National University of Computer and Emerging Sciences
Lahore Campus

Linear Algebra (MT1004)

Date: 21st February 2026

Course Instructors

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Sessional-I Exam

Total Time: 1 Hour

Total Marks: 30

Total Questions: 03

24L-2606
Roll No

BDS-4A
Section


Student Signature

CLO # 1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.

Question # 1 [10 marks] The given matrix represents an augmented matrix for a linear system.

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -4 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}$$

- Write the corresponding set of linear equations for the non homogeneous system.
- Use Gaussian elimination to solve the linear system.

Question # 2 [10 marks] Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$

- Find $|A|$ and state whether the matrix is singular or non-singular
- Find A^{-1} and solve

$$\begin{aligned} x + y + 2z &= 4 \\ 2x + 3y + 5z &= 9 \\ x + z &= 2 \end{aligned}$$

Question # 3 [10 marks]

Suppose that the matrices below are augmented matrices for linear systems in the unknowns x_1, x_2, x_3 and x_4 . These matrices are all in row echelon form but not reduced row echelon form. Discuss the existence and uniqueness of solutions to the corresponding linear systems

$$\begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$